

MADHAV KUMAR YADAV

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PROFESSIONAL SUMMARY

Results-driven AI/ML Python Developer and final-year B.Tech (CSE) student with hands-on experience designing, developing, and deploying end-to-end machine learning pipelines. Proficient in Python, Scikit-learn, Pandas, NumPy, and Matplotlib with strong foundations in machine learning algorithms, statistical modeling, and data preprocessing. Experienced in containerizing and exposing ML models via Docker and FastAPI REST APIs. Collaborative team player with strong communication skills, a proactive learning mindset, and growing expertise in Agentic AI, LangChain, and CrewAI frameworks.

TECHNICAL SKILLS

Programming Languages	Python, MySQL
ML & AI Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, SciPy
ML Techniques	Supervised Learning, Classification, Regression, EDA, Feature Engineering, Hyperparameter Tuning, GridSearchCV, Cross-Validation
MLOps & Deployment	Docker, FastAPI, Flask, Streamlit, MLflow, REST APIs, Render
Databases	MongoDB, MySQL
Developer Tools	Git, GitHub, Jupyter Notebook, VS Code, Power BI, Tableau
Exploring	LangChain, CrewAI, Agentic AI, AI Agents, TensorFlow, PyTorch
Soft Skills	Analytical Thinking, Problem-Solving, Communication, Teamwork, Leadership

PROJECTS

Network Security – Phishing & Intrusion Detection | GitHub Jan 2026

Tech Stack: Python, Scikit-learn, Streamlit, MLflow, MongoDB, Docker

- Designed and developed ML classification models in Python achieving **96% accuracy** in detecting phishing websites and malicious network activity through advanced feature engineering and hyperparameter tuning.
- Integrated **MLflow** for experiment tracking across 50+ runs and **MongoDB** for scalable data storage, improving workflow reproducibility by 40%.
- Containerized the full application with **Docker** and deployed on **Streamlit**, enabling batch prediction of 10,000+ records in under 2 seconds with clean, maintainable Python code.

Student Performance Predictor – End-to-End MLOps Pipeline | GitHub | Live Dec 2025

Tech Stack: Python, Scikit-learn, FastAPI, Docker, Pandas, MLflow

- Built a complete, scalable **MLOps pipeline** predicting student outcomes with **88% F1-score**; automated data preprocessing across 15+ features for 5,000+ records using clean, modular Python code.
- Developed and exposed **REST APIs via FastAPI** (latency <100 ms) for real-time inference; containerized using **Docker** for reproducible, production-ready deployment.
- Implemented experiment logging with **MLflow** and a custom logger, reducing inference latency by **30%** (150 ms → 105 ms) through systematic model optimization.

Forest Fire Weather Index (FWI) Prediction | GitHub Oct 2025

Tech Stack: Python, Scikit-learn, Flask, Pandas, Render

- Developed a **Flask** web application predicting the Forest Fire Weather Index with **92% accuracy**, reducing fire risk assessment time by 60% through automated ML inference.
- Performed comprehensive **EDA** on the Algerian Forest Fire dataset (244 instances, 11 features) to identify key fire triggers; optimized Linear Regression using GridSearchCV with 5-fold cross-validation.
- Deployed on **Render**, handling 500+ real-time predictions per day with 99.9% uptime; documented the full development process and API usage for team collaboration.

EDUCATION

Bachelor of Technology in Computer Science 2023 – 2027

Jagannath University Jaipur, Chaksu, Rajasthan, India

Relevant Coursework: Machine Learning, Data Structures & Algorithms, DBMS, Probability & Statistics, Big Data Analytics, Linear Algebra, Calculus, Applied Mathematics

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Power BI Data Analyst – Udemy
- Python Essentials 1 – Cisco Networking Academy